

API Automation — Team Onboarding Guide

This document explains how to set up, configure, and run the **VRGO / Astro GO API automation framework** from scratch.

For framework internals and extending tests, see the root [README.md](#).

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1. Overview

Item	Detail
Purpose	Automated API regression for VRGO / Astro GO services
Stack	Java 17, Maven, REST Assured, TestNG, Allure, Logback
Optional	Playwright (browser-based token recovery)
Default env	test (Maven profile)

Project layout

```
API-Automation-V3/
├─ src/main/java/com/automation/api/
│  └─ auth/           # Token management, browser recovery
│  └─ client/         # REST Assured API clients
│  └─ config/         # Environment loading
│  └─ constants/      # Endpoints, content kinds
│  └─ model/          # Request/response POJOs
├─ src/test/java/com/automation/api/
│  └─ tests/          # TestNG test classes
│  └─ listeners/      # Allure, Excel, suite bootstrap
├─ src/test/resources/
│  └─ environments/   # Per-env URLs, headers, content IDs
│  └─ testng.xml      # Main regression suite
│  └─ testng-*.xml    # Focused suites
├─ secrets/           # Local auth files (gitignored)
└─ scripts/           # Windows helper scripts
```

```
| docs/           # Documentation (this file)
| pom.xml
| README.md
```

2. Repository access

GitLab (primary)

```
https://gitlab.intelligrape.net/astrogo-automation/apiautomation.git
```

Ask your team lead for access to the **astrogo-automation** group on GitLab.

Clone

```
git clone https://gitlab.intelligrape.net/astrogo-automation/apiautomation.git
cd apiautomation
```

GitHub mirrors (optional)

- <https://github.com/modityajadhav-ai/paytvAstrogoapiautomation-v3.git>
- <https://github.com/modityajadhav-ai/paytvAstrogoapiautomation.git>

3. Prerequisites

Requirement	Notes
JDK 17+	JAVA_HOME must point to a JDK install (not JRE only)
Maven	Optional — use included Maven Wrapper (mvnw.cmd / ./mvnw)
Network	VPN / corporate network if required for target VRGO stacks
Playwright	Optional — install via script for automatic browser re-login

Verify Java and Maven Wrapper

Windows (PowerShell):

```
cd D:\API-Automation-V3
$env:JAVA_HOME = "C:\Program Files\Java\jdk-17" # adjust to your JDK path
.\mvnw.cmd -version
```

Linux / macOS:

```
export JAVA_HOME=/path/to/jdk-17
./mvnw -version
```

4. First-time setup

Step 1 — Clone and open the project

Use your IDE (IntelliJ IDEA recommended) or work from the terminal.

Step 2 — Pick an environment

Profile	Properties file	Typical use
test	src/test/resources/environments/test.properties	Default / dev stack
dev	environments/dev.properties	Development
stage	environments/stage.properties	QA / staging
stage2	environments/stage2.properties	Second stage stack
load	environments/load.properties	Load testing stack
prod	environments/prod.properties	Production (restricted)

Most QA work runs against `stage`.

Step 3 — Create local secrets file

Copy the example template:

```
copy secrets\vrgo-auth.local.properties.example secrets\vrgo-auth.stage.local.properties
```

Edit the new file and fill in credentials. See [Authentication](#).

Important: Files matching `secrets/vrgo-auth.*.local.properties` are gitignored. Never commit them.

Step 4 — (Optional) Install Playwright

Enables automatic browser re-login when refresh tokens expire:

```
scripts\install-playwright.bat
```

Step 5 — Run a smoke test

```
scripts\run-tests.bat stage
```

If auth is configured correctly, tests should start and produce reports under `target/`.

5. Authentication

VRGO tests require a **subscriber refresh token**. Some tests (VR Search Proxy guest flows) also need a **guest bearer JWT**.

Per-environment secrets files

Environment	Secrets file	Token cache
-------------	--------------	-------------

test	secrets/vrgo-auth.test.local.properties	vrgo-token-cache-test.json
dev	secrets/vrgo-auth.dev.local.properties	vrgo-token-cache-dev.json
stage	secrets/vrgo-auth.stage.local.properties	vrgo-token-cache-stage.json
stage2	secrets/vrgo-auth.stage2.local.properties	vrgo-token-cache-stage2.json
load	secrets/vrgo-auth.load.local.properties	vrgo-token-cache-load.json
prod	secrets/vrgo-auth.prod.local.properties	vrgo-token-cache-prod.json

Legacy fallback: `secrets/vrgo-auth.local.properties` (still supported).

Required properties

Property	Required	Description
<code>vrgo.refresh.token</code>	Yes	Subscriber <code>refresh_token</code> from Astro ID login
<code>vrgo.auth.username</code>	Recommended	Test account email (browser recovery)
<code>vrgo.auth.password</code>	Recommended	Test account password (browser recovery)
<code>vrgo.auth.browser.recovery.enabled</code>	Optional	true to auto re-login via Playwright
<code>vrgo.search.proxy.guest.bearer.token</code>	Guest tests only	Guest JWT (~5 min TTL)
<code>vrgo.web.basic.auth.username</code>	Rare	Portal HTTP basic auth (some stacks)
<code>vrgo.web.basic.auth.password</code>	Rare	Portal HTTP basic auth (some stacks)

How to obtain tokens

Subscriber refresh token

1. Open the VRGO web app for your target environment (e.g. stage hub).
2. Log in with a **subscriber** test account (not "Browse as Guest").
3. Open browser DevTools → **Network** tab.
4. Find the auth/token response or any API call.
5. Copy the `refresh_token` value (or the long-lived refresh JWT).
6. Paste into `vrgo.refresh.token=` in your secrets file.

Do not use a guest token as `vrgo.refresh.token`. Guest JWTs have `isGuest: true` and subscriber APIs (e.g. `Continue Watch POST`) will return 401.

Guest bearer token

1. On the web app, choose **Continue As Guest**.
2. DevTools → Network → any VRGO API call → **Authorization** header.
3. Copy the value **after** `Bearer`.
4. Paste into `vrgo.search.proxy.guest.bearer.token=`.
5. Re-paste every ~5 minutes (short TTL).

Browser login URLs (per environment)

Environment	Browser URL
test	https://web.vrgo.test.xp.irdeto.com/hubMovies
dev	https://web.vrgo.dev.xp.irdeto.com/hubHome
load	https://web.vrgo.load.xp.irdeto.com/hubHome
stage	https://web.vr.ctrp-stag.stgbpkastro.com/hubHome
stage2	https://web2.vr.ctrp-stag.stgbpkastro.com/hubHome
prod	https://vrptv.ctrp.astro.com.my/hubHome

Token refresh and cache

- Tokens are cached in `vrgo-token-cache-<env>.json` at the project root (gitignored).
- To force a fresh token: delete the cache file and re-run tests.
- Manual refresh:

```
scripts\refresh-token.bat
```

- Reset all auth caches:

```
scripts\reset-auth-cache.bat
```

CI / pipeline variables

Instead of local files, set these in GitLab CI, Jenkins, or GitHub Actions:

Variable	Purpose
VRGO_REFRESH_TOKEN	Generic refresh token
VRGO_REFRESH_TOKEN_STAGE	Per-env refresh token (suffix = env name, upper case)
VRGO_AUTH_USERNAME_STAGE	Browser recovery username
VRGO_AUTH_PASSWORD_STAGE	Browser recovery password
VRGO_X_API_KEY	API key header (required for prod)
VRGO_BEARER_TOKEN	Optional bootstrap bearer
VRGO_GUEST_BEARER_TOKEN	Guest JWT for CI

6. Running tests

Always run from the **project root**.

Full regression suite

```
# Using script (recommended on Windows)
scripts\run-tests.bat stage

# Using Maven directly
.\mvnw.cmd clean test -Pstage
```

Select environment

```
# Maven profile
.\mvnw.cmd clean test -Pdev
.\mvnw.cmd clean test -Pstage
.\mvnw.cmd clean test -Pload

# System property override
.\mvnw.cmd clean test -Denv=stage

# OS environment variable
$env:ENV = "stage"
.\mvnw.cmd clean test
```

Single test class

```
.\mvnw.cmd clean test -Pstage "-Dtest=com.automation.api.tests.ContinueWatch"
```

Custom TestNG suite

```
.\mvnw.cmd test -Pstage "-Dsurefire.suiteXmlFiles=src/test/resources/testng-continue-watch-
scenarios.xml"
```

Helper scripts (Windows)

Script	What it runs
scripts\run-tests.bat [env]	Full testng.xml suite
scripts\run-tests-no-guest.bat [env]	Suite without guest-dependent tests
scripts\run-continue-watch-scenarios.bat [env]	Continue Watch scenarios only
scripts\run-cw-wa-fav-tests.bat [env]	Continue Watch + Watch Again + Favourites
scripts\run-boxset-binge-tests.bat [env]	Boxset binge flow
scripts\run-vr-search-proxy.bat [env]	VR Search Proxy tests
scripts\run-recommendation-proxy.bat [env]	Recommendation Proxy tests
scripts\refresh-token.bat	Refresh subscriber token manually

scripts\reset-auth-cache.bat	Clear token cache files
scripts\install-playwright.bat	Install Chromium for browser recovery
scripts\verify-browser-recovery.bat	Verify subscriber browser recovery
scripts\verify-guest-browser-recovery.bat	Verify guest browser recovery

Pass the environment as the first argument (default: `test`):

```
scripts\run-tests.bat stage
```

Skip Allure HTML (faster runs)

```
.\mvnw.cmd clean test -Pstage -Dallure.report.skip=true
```

7. Test suites and modules

Main suite (`testng.xml`)

Test class	Module
ContentDetail	Content detail API
ContinueWatch	Continue Watch read/write
ContinueWatchScenarios	Ordered CW scenario flows
WatchAgain	Watch Again API
WatchlistFavourite	Watchlist / Favourites
BoxsetBingeWatchScenarios	Boxset binge + CW/WA flows
LearnAction	Learn action API
RecommendationProxy	Recommendation proxy
VRSearchProxy	VR search proxy (includes guest)
LockChannel	Lock channel
LastTunedChannel	Last tuned channel
SearchHistory	Search history
HomescreenMenu	Homescreen menu
HomescreenProxy	Homescreen proxy
ConfigService	Config service
Footer	Footer API

Focused TestNG suites

File	Purpose
testng-continue-watch-scenarios.xml	CW scenarios (ordered)
testng-cw-wa-fav.xml	CW + Watch Again + Favourites
testng-boxset-binge.xml	Boxset binge watch
testng-vr-search-proxy.xml	VR Search Proxy
testng-recommendation-proxy.xml	Recommendation Proxy
testng-no-guest.xml	Full suite, guest tests excluded

Environment-specific content IDs

Content IDs, catalogue headers, and API paths differ per stack. Edit the active file under `src/test/resources/environments/` :

- `vrgo.header.*` — request headers
- `vrgo.content.movie` , `vrgo.content.tvshow` , etc. — layout content IDs
- `vrgo.cw.add.<kind>.content.id` — Continue Watch POST content IDs

Tests skip a content kind when its ID is blank or starts with `REPLACE` .

8. Reports

After `mvn test` , reports are generated automatically.

Report	Location
Allure HTML	<code>target/site/allure-maven/index.html</code>
Allure raw JSON	<code>target/allure-results/</code>
Extent HTML	<code>target/extent-reports/ExtentReport.html</code>
Excel	<code>excel-reports/</code>

View Allure in browser

```
.\mvnw.cmd allure:serve
```

Clean stale Allure results

Old runs can leave renamed test classes in the report. Use:

```
.\mvnw.cmd clean test -Pstage
```

9. CI and scheduled runs

GitHub Actions

- Workflow: `.github/workflows/api-daily.yml`
- Schedule: daily 7:00 AM IST
- Manual trigger: **Actions** → **API Daily Regression** → **Run workflow**

Required secrets: `SMTP_*` , `REPORT_RECIPIENTS` , `VRGO_REFRESH_TOKEN` , `VRGO_X_API_KEY` , etc.

Jenkins

- Pipeline: `Jenkinsfile`
- Requires JDK 17 tool named `JDK17` (or adjust the `tools` block)
- Email Extension plugin for report emails

Local Windows scheduler

```
copy scripts\.env.example scripts\.env
# Edit scripts\.env with SMTP and recipient settings
.\scripts\register-daily-task.ps1
```

10. Troubleshooting

Symptom	Likely cause	Fix
mvn not recognized	Maven not installed	Use <code>.\mvnw.cmd</code> from project root
release version 17 not supported	Wrong Java version	Set <code>JAVA_HOME</code> to JDK 17
No VRGO refresh token configured	Missing secrets file	Create <code>secrets/vrgo-auth.<env>.local.properties</code>
401 on subscriber APIs	Guest token used as refresh token	Use Astro ID <code>refresh_token</code> , not guest JWT
Guest tests fail after ~5 min	Guest JWT expired	Re-paste <code>vrgo.search.proxy.guest.bearer.token</code>
Browser recovery fails	Playwright not installed	Run <code>scripts\install-playwright.bat</code>
Device limit errors	Too many registered devices	Set <code>vrgo.auth.browser.evict.device.on.limit=true</code>
Old test names in Allure	Stale results	Run <code>mvn clean test</code>
PowerShell -d parsing errors	Shell eats -d flags	Quote JVM args: <code>"-Dtest=..."</code>

11. Security rules

1. **Never commit** `secrets/vrgo-auth.*.local.properties` , token cache JSON, or `scripts/.env` .

2. Share credentials only via approved secure storage (1Password, Vault, GitLab masked variables).
 3. Do **not** paste tokens or passwords in Slack, email, or tickets.
 4. Rotate tokens immediately if accidentally committed or exposed.
 5. Use dedicated test accounts — not personal production accounts.
 6. Production (`prod` profile) requires extra approval; never hardcode prod secrets in source files.
-

Quick start checklist

- ☐ GitLab access granted
 - ☐ Repo cloned
 - ☐ JDK 17 installed, `JAVA_HOME` set
 - ☐ `secrets/vrigo-auth.stage.local.properties` created from example template
 - ☐ Subscriber refresh token and test account credentials added
 - ☐ `scripts\run-tests.bat` stage runs successfully
 - ☐ Allure report opens at `target/site/allure-maven/index.html`
 - ☐ (Optional) Playwright installed for browser recovery
-

Getting help

- Framework details: [README.md](#)
- Manual test cases: `docs/qa/manual-testcases/`
- Boxset binge test design: `docs/qa/boxset-binge-watch-cw-wa-test-suite.md`
- Contact: your team's QA / automation lead